Weekly Radar
Anomaly Detection on Weekly Network Performance

Project 06 - Telco-Cloud Analytics Training

Information Technology Institute

Team Members

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Dataset: Performance.xlsx → weekly_region_carrier_tech.csv
Date: April 2026

Dataset & Filtering

What was the dataset?

The source workbook was a **mobile network performance dataset** with **93,364 rows**. It mixed **daily and weekly records**, contained both **locality-level rows** and a **country aggregate**, and covered multiple **regions**, **carriers**, and **technologies**.

1) Removed daily rows

We kept **weekly rows only** so that all comparisons used the same time granularity.

2) Kept locality rows

The main workflow used **weekly locality rows** because they are the correct starting level for aggregation.

3) Separated country rows

Country aggregate rows were moved out of the main analysis table and kept only as benchmark rows.

4) Aggregated by region

Locality rows were aggregated into a clean **region × carrier × technology × week** table using **sample_count-weighted averages**.

5) Applied a sample threshold

We removed weak groups with very small sample sizes to make the final analysis more reliable.

6) Added helper columns

Helper columns were added to support **mean vs. median** comparison and later anomaly analysis.

Preprocessing pipeline results

The table summarizes how the raw workbook was filtered and reduced into the final analysis dataset.

Processing stage	Rows
Raw spreadsheet rows	93,364
Weekly rows kept	58,858
Daily rows removed	34,506
Weekly locality rows	58,669
Weekly country rows	189
Aggregated region-carrier-technology-week rows	351
Rows removed by sample threshold	16
Final analysis rows	**335**

Table 1: Row counts through the preprocessing pipeline.

EDA Highlights

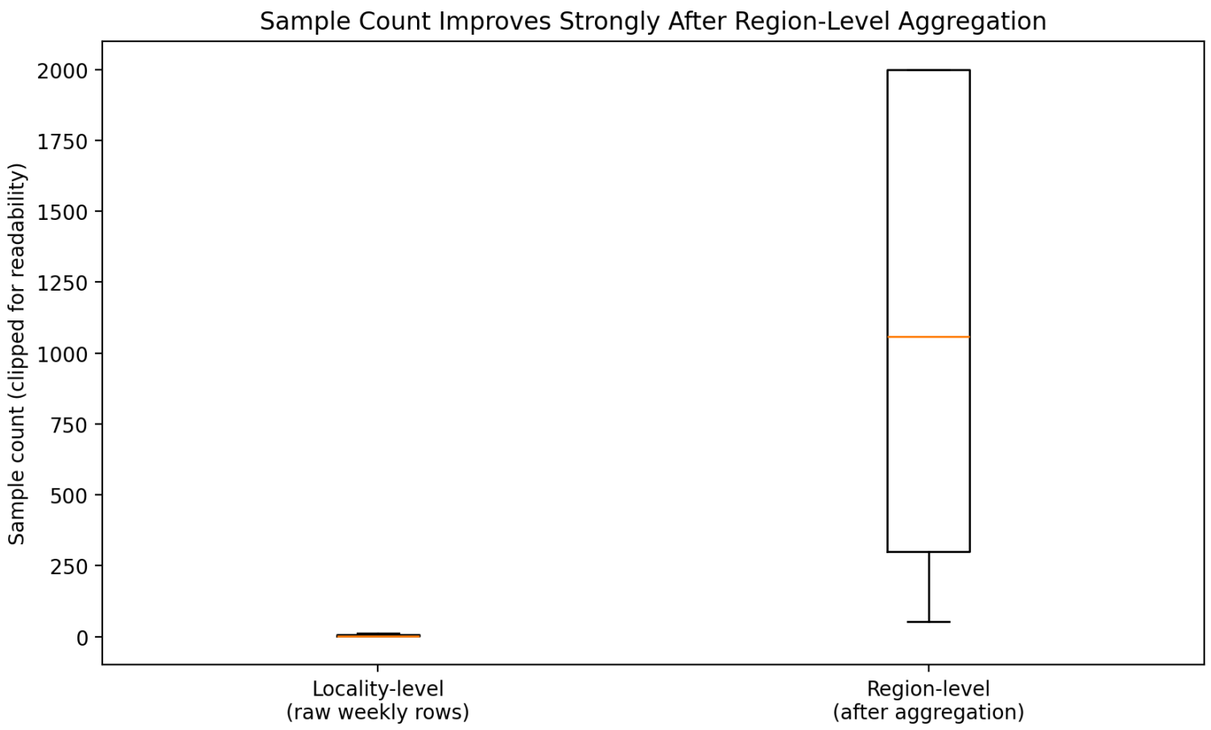

Figure 1: Sample-count strength before and after region-level aggregation.

Main EDA takeaway

- **Aggregation greatly improved data reliability.**
- **Locality-level rows** had weak sample support and were too noisy for fair comparison.
- The aggregated **region-level table** produced much stronger and more stable records.
- This made the cleaned dataset suitable for **peer comparison** and later anomaly analysis.

Carrier-wise technology comparison

Across carriers, the same broad technology pattern appeared: 5G delivered the strongest download levels, **Multi-RAT** stayed in the middle, and **LTE** was generally the weakest. This suggests that the technology effect was **consistent across carriers**, not driven by only one operator.

Q1 Result - Which carrier is consistently below the regional average?

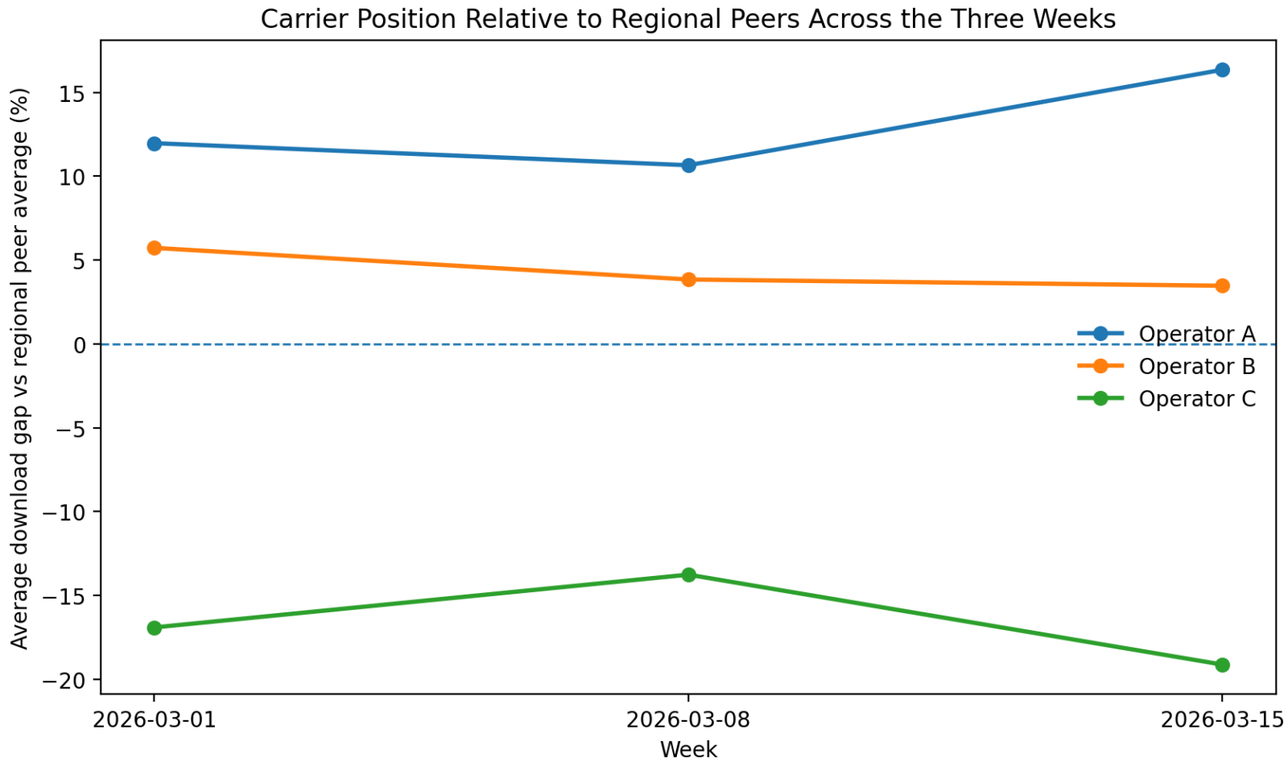

Figure 2: Average carrier download gap relative to regional peers across the three weeks.

Answer to Q1

Operator C was the only carrier that stayed **below the regional peer average in all three weeks**. Operators A and B remained above the peer baseline on average.

Interpretation

This is not a one-week spike. It indicates a repeated structural weakness relative to peer carriers.

How this graph was produced

- For each **week**, we computed each carrier's **average download** inside every region.
- We then calculated the **regional peer average** in the same week using the other carriers as the benchmark.
- The plotted metric is the **gap (%) versus the peer average**: positive means above peers, negative means below peers.

Q2 Result - Which regions show the largest week-over-week changes?

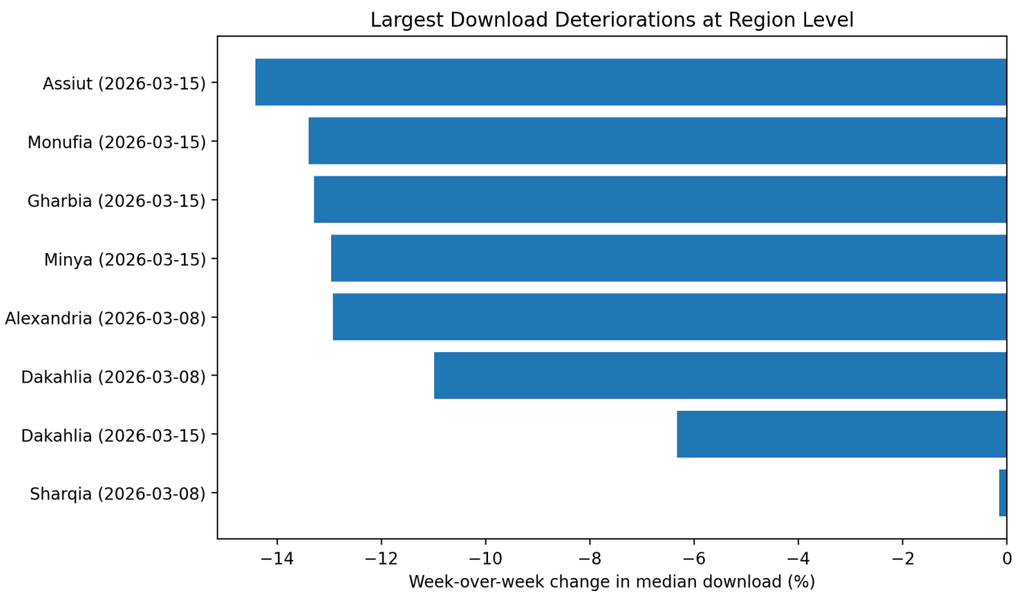

Figure 3: Largest week-over-week deteriorations in region-level median download.

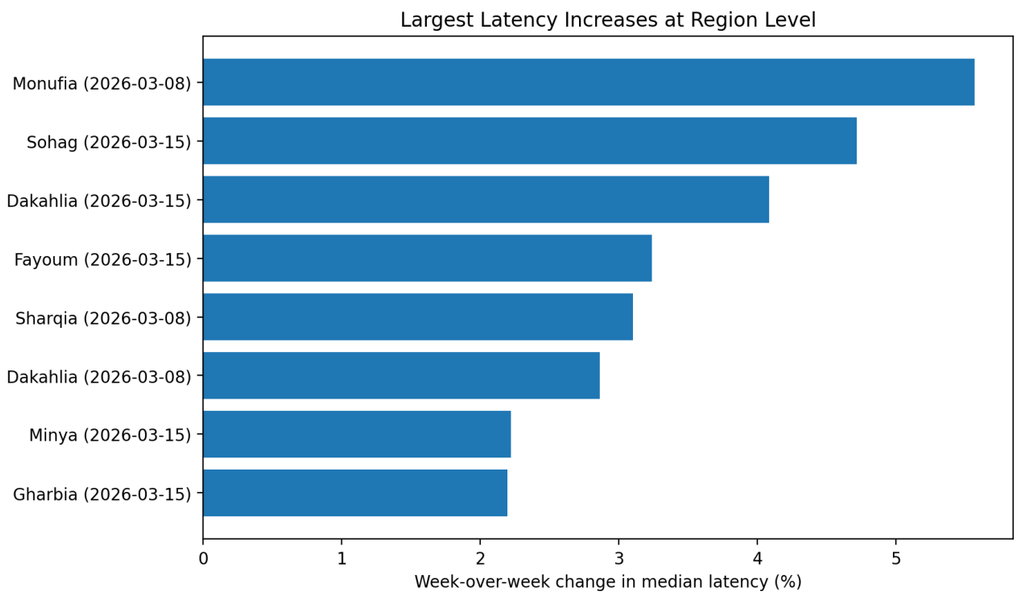

Figure 4: Largest week-over-week increases in region-level median latency.

Answer to Q2

The sharpest **download deteriorations** appeared in **Assiut (-14.42%)**, **Monufia (-13.39%)**, and **Gharbia (-13.29%)**. The strongest **latency increases** appeared in **Monufia (+5.57%)**, **Sohag (+4.71%)**, and **Dakahlia (+4.08%)**.

Week-over-week percentage change equation

$$\text{KPI WoW (\%)} = \frac{\text{Current Week Median KPI} - \text{Previous Week Median KPI}}{\text{Previous Week Median KPI}} \times 100$$

Q3 Result - Which region-carrier pairs are peer outliers?

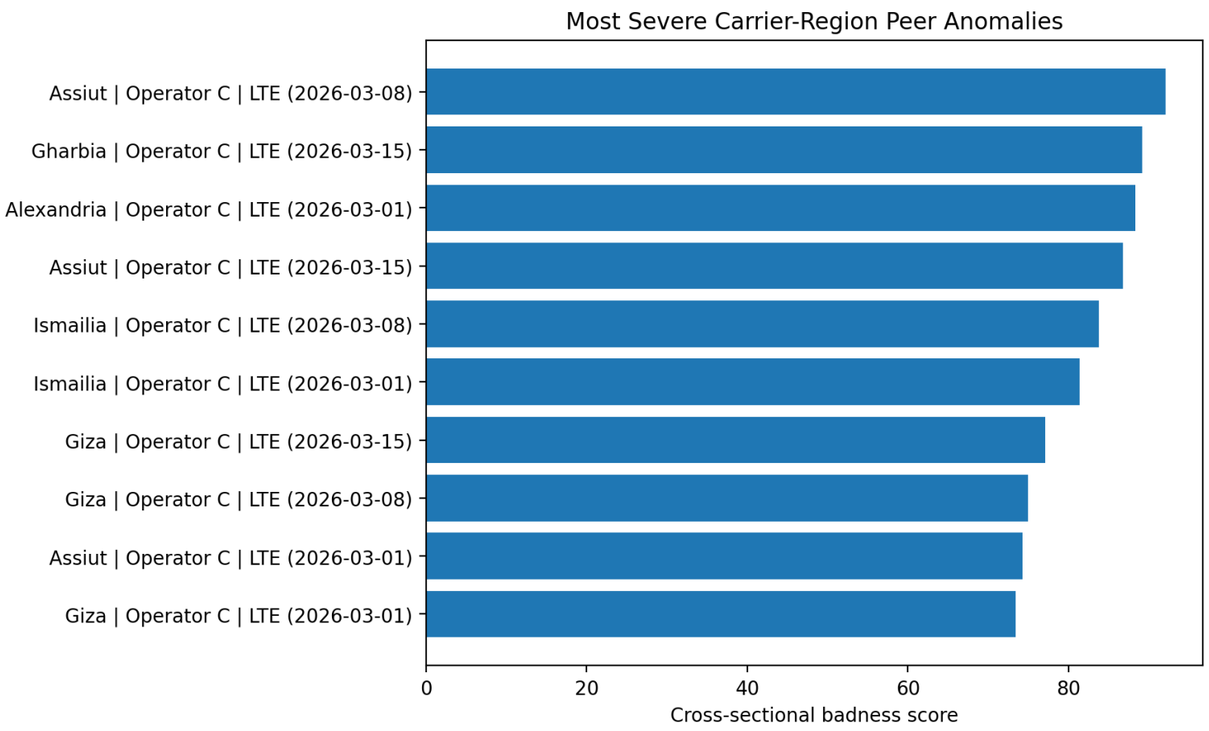

Figure 5: Most severe carrier-region peer anomalies from the cross-sectional badness score.

Answer to Q3

Operator C on LTE dominated the strongest peer anomalies, especially in **Assiut, Gharbia, Alexandria, Ismailia,** and **Giza**. These cases combined much lower download with higher latency than peer carriers in the same region and week.

How this graph was produced

- Each **region-carrier-week-technology** record was compared against the **other carriers** in the same region, week, and technology.
- We computed a combined peer-anomaly score that penalizes **lower download** and **higher latency**, then ranked the highest scores.

$$\text{Cross-sectional Badness} = \max(0, -\text{Download Gap (\%)}) + \max(0, \text{Latency Gap (\%)})$$

Q4 & Q5 Results - Technology Comparison and Region Ranking

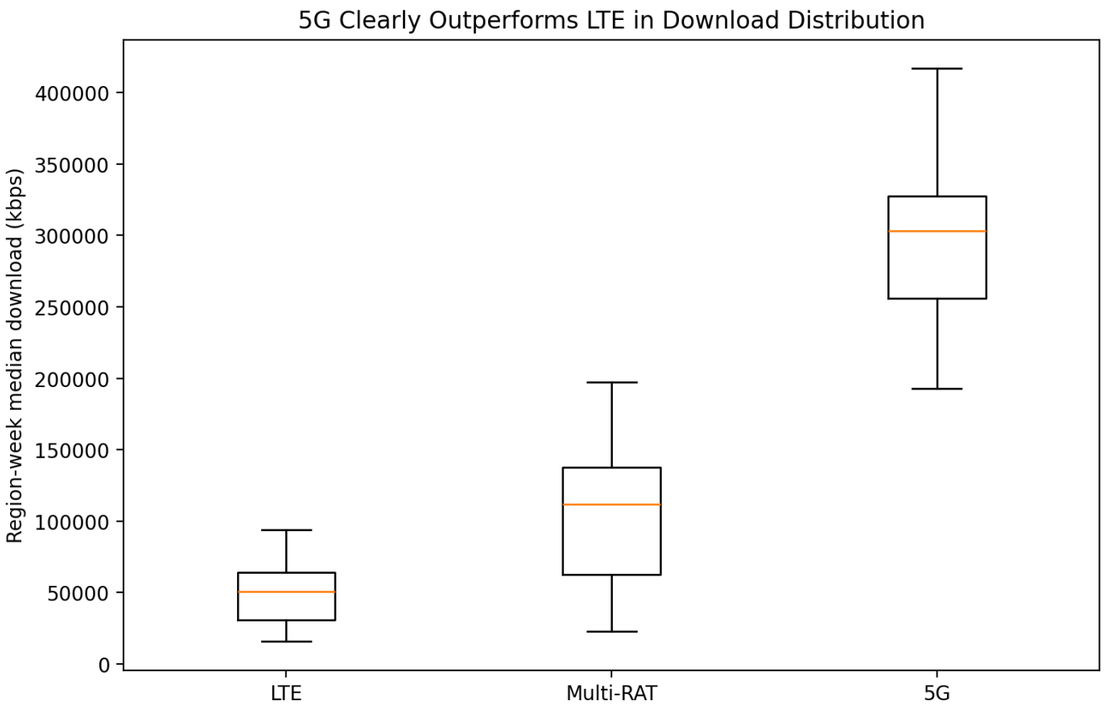

Figure 6: Technology-wise region-week download distribution.

Figure 7: Region ranking by week (1 = best).

Answers

- **Q4:** Across the **39 valid region-week comparisons**, 5G download was higher than LTE in **39/39 cases**; average region-week median download was about **296 Mbps for 5G** versus **49 Mbps for LTE**.
- **Q5:** Ranking was built from **sample-count-weighted regional median download**; strongest regions were **Port Said, Gharbia,** and **Assiut**.
- **Stability: Port Said** had the largest movement (**1 → 3 → 1**), while **Sohag** stayed last (**13 → 13 → 13**).

Methods (P2 & P3) - How the anomaly detection was built

Method	What it does	Supports
Cross-sectional peer comparison	Compares each carrier against the regional peer average inside the same week and technology.	Q1, Q3
Week-over-week change analysis	Measures percentage change between consecutive weeks at region level to detect deterioration or improvement.	Q2
Technology distribution comparison	Compares LTE, Multi-RAT, and 5G download distributions across cleaned region-week records.	Q4
Region ranking analysis	Ranks all regions by region-week median download and tracks rank changes across the three weeks.	Q5

Table 2: Method design used to complete Project 06.

Why this method is appropriate

The dataset contains only **three weekly time points**, so long-history temporal models are not the best first choice. Cross-sectional comparison is more suitable because it compares carriers and regions against direct peers at the same time.

Most Surprising Finding

Top Cross-Sectional Anomaly Scores

Week	Region	Carrier	Tech	DL Gap	Lat Gap	Score
2026-03-08	Assiut	Operator C	LTE	-67.5%	70.6%	1.983
2026-03-15	Assiut	Operator C	LTE	-68.5%	66.8%	1.974
2026-03-01	Alexandria	Operator C	LTE	-65.2%	57.9%	1.905
2026-03-08	Ismailia	Operator C	LTE	-61.7%	69.1%	1.905
2026-03-15	Giza	Operator C	LTE	-65.8%	42.3%	1.897
2026-03-01	Ismailia	Operator C	LTE	-61.9%	64.4%	1.897
2026-03-01	Assiut	Operator C	LTE	-52.0%	84.0%	1.871
2026-03-08	Giza	Operator C	LTE	-65.3%	40.2%	1.871
2026-03-01	Monufia	Operator C	Multi-RAT	-52.7%	37.7%	1.855
2026-03-15	Gharbia	Operator C	LTE	-56.6%	63.2%	1.853
2026-03-01	Cairo	Operator B	Multi-RAT	-51.2%	34.2%	1.838
2026-03-01	Giza	Operator C	LTE	-66.1%	33.3%	1.836
2026-03-08	Monufia	Operator C	Multi-RAT	-65.4%	22.5%	1.829
2026-03-15	Assiut	Operator C	5G	-24.7%	67.9%	1.818
2026-03-15	Ismailia	Operator C	LTE	-51.1%	57.5%	1.802

Table 3: Highest anomaly scores from the peer-comparison anomaly log.

Most surprising result

Operator C dominated the anomaly detection results, especially on **LTE**.

- **9 of the top 10** peer anomalies were driven by **Operator C**, and most were **LTE** cases.
- The strongest C cases were not mild deviations: **download was 52–68% below peers** while **latency was 33–84% above peers**.
- The issue repeated across several regions: **Assiut, Alexandria, Ismailia, Giza, Gharbia,** and **Monufia**.
- **Operator B also appeared** in high-severity Multi-RAT anomalies, especially **Cairo** and **Dakahlia**, proving the log is not blindly carrier-biased.

Interpretation: Cross-sectional scoring exposed a persistent carrier-technology weakness rather than a single regional accident.

Lessons Learned

1) Preprocessing is part of the model

The biggest improvement came before anomaly detection: removing daily rows, separating country aggregates, and aggregating localities with **sample-count weighting**.

2) Cross-sectional comparison was the right fit

With only **3 weekly snapshots**, comparing each carrier against **same-week, same-region, same-technology peers** gives stronger evidence than forcing a time-series model.

3) Why not long-history models first?

- Rolling z-score / EWMA needs many historical weeks.
- LSTM or forecasting models need enough history to learn normal behavior.
- Example: with only 3 weeks, a sudden drop cannot be separated from normal seasonality.

4) Interpretability matters

A good anomaly record should answer: **which region, which carrier, which technology, which week, and which KPI gap** caused the alert.

Key technical takeaway

Cross-sectional analysis is strongest when the dataset has **few time points but many peer entities**. Here, we had only 3 weeks but many comparable region-carrier-technology records, so peer comparison was more reliable than long-window temporal modeling. With 6--12+ months of weekly data, the next step would be adding rolling baselines, CUSUM/EWMA, or temporal anomaly models and validating them against known network incidents.